

Documentação de criação da rede local

Philippe Siqueira Ferreira

Atividade

Apresentação do trabalho para amanhã.

1) apresentar um cenário com uma infraestrutura de rede com os seguintes características:

A) preparar uma rede utilizando o SW com 12 dispositivos em VLAN, sabendo 06 portas para diretoria e 06 para o executivo.

B) no mesmo SW as demais 12 portas devem estar inacessível.

C) colocar senha no SW.

D) colocar senha na porta console.

E) colocar senha no acesso virtual.

F) criptografar a senha.

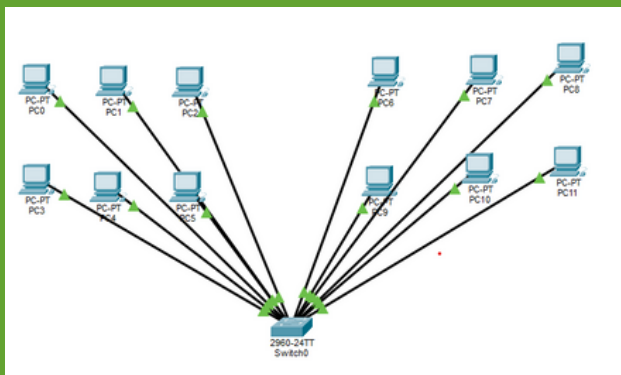
G) configurar o SW que na tentativa de acesso o mesmo apresente a mensagem de alerta aviso a critério do aluno.

Introdução

Venho por este documento pontuar as principais ações realizadas para a criação da rede, com o fim de facilitar futuras melhorias e manutenções necessárias. Por assim, esse documento é necessário para melhorar a manutenibilidade do sistema

Ferramentas e equipamentos utilizados

1. Ferramenta: Cisco Packet Tracer;
2. 1 Switch Cisco 2960;
3. 12 Computadores (6 para cada Vlan)



Foi montada duas redes vlan para a operacao. Com a utilizacao de um total de 12 computadores. 6 para cada uma das redes vlan.

As portas utilizadas para as operacoes foram as respectivas a baixo para cada uma das maquinas no switch 2960:

FastEthernet 0/1 → PC - PT PC0 (DIRETORIA)
FastEthernet 0/2 → PC - PT PC1 (DIRETORIA)
FastEthernet 0/3 → PC - PT PC2 (DIRETORIA)
FastEthernet 0/4 → PC - PT PC3 (DIRETORIA)
FastEthernet 0/5 → PC - PT PC4 (DIRETORIA)
FastEthernet 0/6 → PC - PT PC5 (DIRETORIA)
FastEthernet 0/7 → PC - PT PC6 (EXECUTIVO)
FastEthernet 0/8 → PC - PT PC7 (EXECUTIVO)
FastEthernet 0/9 → PC - PT PC8 (EXECUTIVO)
FastEthernet 0/10 → PC-PT PC10 (EXECUTIVO)
FastEthernet 0/11 → PC-PT PC11 (EXECUTIVO)
FastEthernet 0/12 → PC-PT PC12 (EXECUTIVO)

Logo após criar a conexão das máquinas com as conexões com o switch, no CLI é mostrado um informativo das conexões realizadas

```
Press RETURN to get started!
```

```
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/7, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/7, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/8, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/8, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/9, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/9, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/10, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/11, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/11, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/12, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/12, changed state to up
!
```

Após realizar as conexões é necessário criar as duas redes VLAN e em seguida dedicar cada máquina para cada cenário

Os comandos utilizados foram os seguintes:

vlan 10 → Cria uma vlan com o numero identificador 10;

name DIRETORIA → A vlan criada será nomeada diretoria;

```
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name DIRETORIA
Switch(config-vlan)#exit
Switch(config)#
```

Mesma logica de antes é aplicada, mudando apenas o nome e o numero utilizado

```
Switch(config)#vlan 20
Switch(config-vlan)#name EXECUTIVO
Switch(config-vlan)#exit
Switch(config)#
```

VLAN Name		Status	Ports							
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2							
10	DIRETORIA	active								
20	EXECUTIVO	active								
1002	fdi-default	active								
1003	token-ring-default	active								
1004	fdinet-default	active								
1005	trnet-default	active								
VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	0	0
10	enet	100010	1500	-	-	-	-	-	0	0
20	enet	100020	1500	-	-	-	-	-	0	0
1002	fdi	101002	1500	-	-	-	-	-	0	0
1003	tr	101003	1500	-	-	-	-	-	0	0
1004	fdnet	101004	1500	-	-	-	ieee	-	0	0
1005	trnet	101005	1500	-	-	-	ibm	-	0	0
VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
Remote SPAN VLANs										
Primary	Secondary	Type	Ports							
Switch#										

Em seguida utilizo o comando **show vlan** para mostrar as vlans existentes e as portas disponíveis em cada cenário

```

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface range fastEthernet 0/1 - 6
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

```

O comando **interface range fastEthernet 0/1 - 6** faz com que as portas de 1 a 6 do switch sejam selecionadas;

Em seguida o comando **switchport mode access** seleciona as portas para serem acessadas, e o comando é complementado com o comando **switchport access vlan 10** concedendo o acesso dessas 6 portas na VLAN 10

show vlan												
VLAN Name				Status	Ports							
1	default			active	Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/1, Gig0/2 Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6							
10	DIRETORIA			active								
20	EXECUTIVO			active								
1002	fddi-default			active								
1003	token-ring-default			active								
1004	fddinet-default			active								
1005	trnet-default			active								
VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2		
1	enet	100001	1500	-	-	-	-	-	0	0		
10	enet	100010	1500	-	-	-	-	-	0	0		
20	enet	100020	1500	-	-	-	-	-	0	0		
1002	fddi	101002	1500	-	-	-	-	-	0	0		
1003	tr	101003	1500	-	-	-	-	-	0	0		
1004	fdnet	101004	1500	-	-	-	ieee	-	0	0		
1005	trnet	101005	1500	-	-	-	ibm	-	0	0		
VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2		
Remote SPAN VLANs												
Primary	Secondary	Type	Ports									

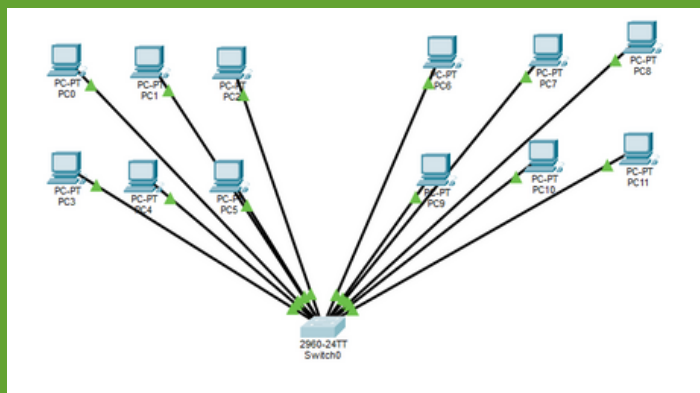
O comando **show vlan** confirma a situação das 6 portas serem alocadas para a VLAN diretoria


```

Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface range fastEthernet 0/1 - 12
Switch(config-if-range)#switchport mode access
^
% Invalid input detected at '^' marker.

Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport port-security
Switch(config-if-range)#switchport port-security maximum 1
Switch(config-if-range)#switchport port-security violation shutdown
Switch(config-if-range)#exit
Switch(config)#

```



```
Switch#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
      (Count)      (Count)      (Count)
-----
Fa0/1      1          0          0          Shutdown
Fa0/2      1          0          0          Shutdown
Fa0/3      1          0          0          Shutdown
Fa0/4      1          0          0          Shutdown
Fa0/5      1          0          0          Shutdown
Fa0/6      1          0          0          Shutdown
Fa0/7      1          0          0          Shutdown
Fa0/8      1          0          0          Shutdown
Fa0/9      1          0          0          Shutdown
Fa0/10     1          0          0          Shutdown
Fa0/11     1          0          0          Shutdown
Fa0/12     1          0          0          Shutdown
-----
```

```
Switch#show interface status
Port      Name      Status      Vlan      Duplex      Speed      Type
Fa0/1     Fa0/1     connected   10        a-full      a-100     10/100BaseTX
Fa0/2     Fa0/2     connected   10        a-full      a-100     10/100BaseTX
Fa0/3     Fa0/3     connected   10        a-full      a-100     10/100BaseTX
Fa0/4     Fa0/4     connected   10        a-full      a-100     10/100BaseTX
Fa0/5     Fa0/5     connected   10        a-full      a-100     10/100BaseTX
Fa0/6     Fa0/6     connected   10        a-full      a-100     10/100BaseTX
Fa0/7     Fa0/7     connected   20        a-full      a-100     10/100BaseTX
Fa0/8     Fa0/8     connected   20        a-full      a-100     10/100BaseTX
Fa0/9     Fa0/9     connected   20        a-full      a-100     10/100BaseTX
Fa0/10    Fa0/10    connected   20        a-full      a-100     10/100BaseTX
Fa0/11    Fa0/11    connected   20        a-full      a-100     10/100BaseTX
Fa0/12    Fa0/12    connected   20        a-full      a-100     10/100BaseTX
Fa0/13    Fa0/13    notconnect  1         auto        auto      10/100BaseTX
Fa0/14    Fa0/14    notconnect  1         auto        auto      10/100BaseTX
Fa0/15    Fa0/15    notconnect  1         auto        auto      10/100BaseTX
Fa0/16    Fa0/16    notconnect  1         auto        auto      10/100BaseTX
Fa0/17    Fa0/17    notconnect  1         auto        auto      10/100BaseTX
Fa0/18    Fa0/18    notconnect  1         auto        auto      10/100BaseTX
Fa0/19    Fa0/19    notconnect  1         auto        auto      10/100BaseTX
Fa0/20    Fa0/20    notconnect  1         auto        auto      10/100BaseTX
Fa0/21    Fa0/21    notconnect  1         auto        auto      10/100BaseTX
--More--
```

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#enable secret cisco123
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
exit
```

```
Switch>enable
Password:
Switch#
```

```
Switch>enable
Password:
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#line console 0
Switch(config-line)#password console1
Switch(config-line)#login
Switch(config-line)#exit
Switch(config)#
```

```
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
exit
```

Switch con0 is now available

Press RETURN to get started.

User Access Verification

Password:

Switch>

```
Password:
Switch>enable
Password:
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#line vty 0 4
Switch(config-line)#password vty123
Switch(config-line)#login
Switch(config-line)#exit
Switch(config)#
```

```
Switch(config)#service password-encryption
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
Switch#S
```

```
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#banner motd #
Enter TEXT message. End with the character '#'.
#
* AVISO DE SEGURANA DO ENGENHEIRO PHILIPHE
*
* Acesso restrito somente a usuarios
* autorizados.
* Tentativas no autorizadas zero
* monitoradas e registradas.
#
Switch(config)#
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
```

Switch#exit

Switch con0 is now available

Press RETURN to get started.

```
*****
*   AVISO DE SEGURANA DO ENGENHEIRO PHILIPHE   *
*                                                                 *
*   Acesso restrito somente a usuarios         *
*   autorizados.                               *
*   Tentativas no autorizadas sero             *
*   monitoradas e registradas.                 *
*****
```

|
User Access Verification

Password:
% Password: timeout expired!

